BRIGHTLOOP

Subscription Control Tower

Hackathon product plan: complete subscription management + Digital Twin Agentic AI

Core promise Manage the complete subscription lifecycle from quotation to payment.	**Novel differentiator** Simulate pricing, discounts, renewals, and revenue safely before changing live data.

The baseline follows the Brightloop problem statement: products, plans, subscriptions, quotations, invoices, taxes, discounts, payments, overdue tracking, role-based access, and finance reporting.

Prepared for hackathon implementation

1. Product vision

Brightloop Subscription Control Tower is a unified platform for running a subscription business. It combines an operational billing system with a safe simulation layer that helps finance teams make better decisions.

“We do not only record what happened. We help the business understand what will happen next.”

The core system satisfies the required workflow. The Digital Twin Agentic AI is an additional feature that turns the system into an intelligent, explainable revenue-control platform.

2. Core features from the brief

Area	Required capability
Users & access	Admin, internal staff, and customer roles; unique email; password rules; reset link; customer-only portal access.
Products & plans	Products, variants, sales/cost prices, notes, billing frequency, minimum quantity, availability dates, pause/renew/close rules.
Quotation flow	Quotation templates with product lines, plans, payment terms, and validity period.
Subscription lifecycle	Draft -> Quotation -> Confirmed -> Active -> Closed, with number, customer, plan, dates, lines, taxes, and totals.
Automatic billing	Generate invoices from subscriptions; calculate taxes; prevent duplicate or skipped billing.
Discounts	Fixed or percentage rules, minimum quantity, dates, usage limits, product or whole-subscription scope.
Payments	Record method, amount, date, note; match payments to invoices; track paid, partial, outstanding, and overdue.
Finance dashboard	Active subscriptions, revenue, payments, overdue invoices, filters, summary cards, and fast response at scale.

3. Novel feature: Digital Twin Agentic AI

The Digital Twin is an isolated, simulated copy of the Brightloop business. It mirrors customers, products, plans, subscriptions, discounts, invoices, payments, and renewal dates, but it cannot change live records during a simulation.

What the agent can answer

• What happens if we increase the monthly plan price by 10%?

• Which subscriptions are most likely to become overdue next month?

• What if all discounts expiring this month are removed?

• Which customers should receive an upgrade offer?

• How much revenue is at risk next month?

The agent returns

• Predicted revenue and revenue at risk

• Affected subscriptions and customers

• Expected overdue amount

• Discount and tax impact

• Customer-risk indicators

• A plain-language recommendation

4. Agent workflow and safety

Step	What happens
1. Understand	The local AI interprets the finance lead question.
2. Select tool	It chooses an approved simulation function.
3. Calculate	Python performs exact revenue, tax, discount, and invoice calculations.
4. Explain	The AI summarizes the result and its assumptions.
5. Approve	The user reviews a proposed change.
6. Apply	Only after approval does the system update real Odoo data.

5. Standout supporting features

Billing-run simulator

• Preview invoice count, total revenue, taxes, discounts, missing data, and duplicate-billing risks before creating invoices.

Explainable invoices

• Show how each invoice total was formed: plan price, variant charge, quantity, discount, taxes, and final amount.

Subscription state guard

• Block invalid transitions such as activating an unconfirmed subscription, billing a closed subscription, or applying an expired discount.

Payment matching assistant

• Suggest invoice matches using customer, amount, date, method, and note; handle partial and over-payments.

6. Technical architecture

Layer	Implementation
Core business system	Odoo Community Edition or an Odoo-style Python application.
Data	PostgreSQL for the main system; SQLite is acceptable for a small demo.
Digital Twin	A separate snapshot of business records used only for simulation.
Simulation engine	Python functions for pricing, tax, discount, revenue, payment, and risk calculations.
Agent	Local Ollama model such as Qwen, Llama, or Mistral.
Interface	Odoo views or a Streamlit finance/AI dashboard.
Safety	Tool allow-list, audit log, and approval before every real-data change.

Free-cost AI plan

• Run the model locally with Ollama.

• Use Python for all exact financial calculations.

• Send only the user's question and compact results to the local model.

• Avoid paid APIs and keep business data private.

7. Hackathon demo storyline

• Admin creates a product, variant, tax rule, and monthly plan.

• Sales staff creates a quotation from a template.

• The quote becomes a confirmed, active subscription.

• Billing day automatically creates an invoice with taxes and discount.

• A payment is recorded and matched to the invoice.

• A second invoice becomes overdue and appears on the finance dashboard.

• The finance lead asks: "What happens if we increase the monthly plan by 10%?"

• The Digital Twin shows revenue change, affected customers, risk, and recommendation.

• The finance lead approves or rejects the proposed change.

The judge sees both the required end-to-end workflow and a memorable intelligence layer built directly on top of it.

8. Build priority

Priority	Build
1	Authentication, roles, products, plans, and subscriptions

Priority	Build
2	Quotation-to-active lifecycle
3	Automatic invoices, taxes, discounts, and payments
4	Overdue tracking and finance dashboard
5	Digital Twin snapshot and scenario simulator
6	Local AI agent and approved tool calls
7	Approval workflow, explainable invoices, and billing preview

Final pitch

“Brightloop Subscription Control Tower manages the complete subscription lifecycle—from quotation to payment—and adds a Digital Twin Agent that lets finance teams safely simulate pricing, discounts, renewals, and revenue decisions before changing the real business.”